

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – IV
PAPER –2
TEST DATE: 31-07-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

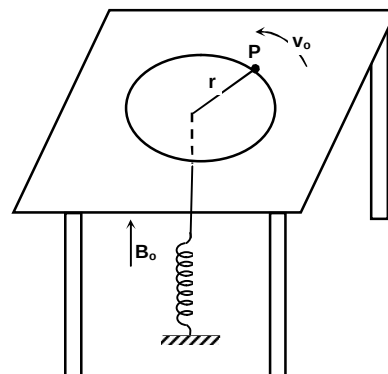
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Two projectile are thrown at the same time from two different points. The projectile thrown from the origin has initial velocity $3\hat{i} + 3\hat{j}$ with respect to earth. The projectile has initial velocity $a\hat{i} + b\hat{j}$ with respect to earth thrown from the point (10, 5). ($\hat{i}$ is a unit vector along horizontal, $\hat{j}$ along vertical). If the projectile collides after two second, then the

- (A) value of a is -2
- (B) value of a is $\frac{1}{2}$
- (C) value of b is $\frac{1}{2}$
- (D) value of b is -2

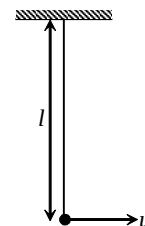
2. A particle mass m charge q is moving on a circular path on the surface of a frictionless table with speed v_0 where a magnetic field B_0 exists uniformly over the whole region. It is attached by a string which passes through a hole in the table to a spring as shown.



The spring is stretched by x_0 . If now the magnetic field is increased slowly to $2B_0$, which of these are incorrect?

- (A) The extension in the spring will increase
- (B) The speed of the particle will increase
- (C) The speed of the particle will decrease
- (D) The kinetic energy of the particle will decrease

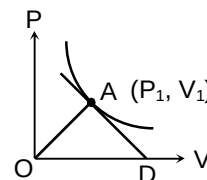
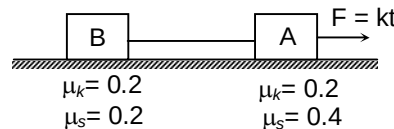
3. A particle of mass m is attached to one end of the light inextensible string and other end of the string is fixed in vertical plane as shown.



Now particle is given horizontal velocity $u = \sqrt{\frac{5}{2}gl}$, then which of the following statement/s is/are correct.

- (A) The string makes angle with the downward vertical is 60° when acceleration of the particle is horizontal.
- (B) The tension in string is $2mg$ when acceleration of the particle is horizontal.
- (C) The speed of the particle is $\sqrt{\frac{3}{2}gl}$ when acceleration of the particle is horizontal
- (D) The particle will not complete the circle.

4. Two block A and B each of mass $\frac{1}{2}$ kg is connected by a massless inextensible string and kept on horizontal surface. Coefficient of friction between block and surface shown in figure. A force $F = kt$ (where $k = 1$ N/s and t is time in second) applied on A. Then ($g = 10$ m/s²)
- (A) work done by friction force on block B is zero in time interval $t = 0$ to $t = 3$ s.
 (B) work done by friction force on block A is zero in time interval $t = 0$ to $t = 3$ s.
 (C) work done by tension on B is also zero in time interval $t = 0$ to $t = 3$ s.
 (D) speed of blocks at $t = 10$ s is 27.5 m/s.
5. A traveling wave pulse is given by $y = \frac{6}{2 + (x + 3t)^2}$ where symbols have their usual meanings, x, y are in metre and t is in second. Then
- (A) the pulse is traveling along +ve x -axis with velocity 3 m/s
 (B) the pulse is traveling along -ve x -axis with velocity 3 m/s
 (C) the amplitude of the wave pulse is 3 m
 (D) the pulse is a symmetric pulse
6. n moles of an ideal gas undergo an isothermal process at temperature T . P-V graph of the process is as shown in the figure. A point A (V_1, P_1) is located on the P-V curve. Tangent at point A, cuts the V-axis at point D. AO is the line joining the point A to the origin O of PV diagram. Then,
- (A) coordinates of points D is $\left(\frac{3V_1}{2}, 0\right)$
 (B) coordinates of points D is $(2V_1, 0)$
 (C) area of the triangle AOD is nRT
 (D) area of the triangle AOD is $\frac{3}{4}nRT$

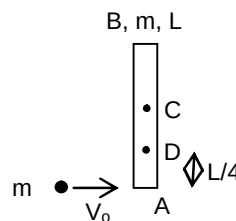


Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

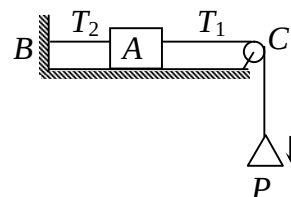
A thin, uniform rod of mass M and length L is at rest on a smooth horizontal surface. A particle of same mass M collides with the rod at one end perpendicular to its length with velocity V_0 and sticks to it. C is the middle point of the rod and D is a point at a distance of $L/4$ from end A.



7. Choose the correct statement out of the following.
- (A) Mechanical energy of the particle and rod system is conserved in collision.
 - (B) mechanical energy of the particle and the rod individually is conserved.
 - (C) linear momentum of the particle and the rod are individually conserved.
 - (D) linear momentum of the rod and the particle system is conserved.
8. Velocity of point D of the rod immediately after collision will be
- (A) $\frac{V_0}{4}$
 - (B) $\frac{V_0}{2}$
 - (C) $\frac{3V_0}{4}$
 - (D) V_0

Paragraph for Question Nos. 09 and 10

A block A of mass 1 kg is kept on rough surface of co-efficient of friction 0.5. A light pan P is attached with a light string ACP to block A. The block A is also connected to wall by another thread AB. Sand particles are poured at a constant rate $\frac{dm}{dt} = 0.1 \text{ kg/s}$ into the pan P. The sand particles strike the pan P with 10 m/s and stop. The pouring starts at $t = 0$. ($g = 10 \text{ m/s}^2$)



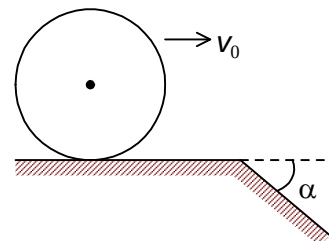
9. The tension T_1 in thread ACP at $t = 5\text{s}$
- (A) 5 N
 - (B) 4 N
 - (C) 6 N
 - (D) 1 N
10. The impulse given by friction force on the block A till $t = 5\text{s}$
- (A) 17 N-s
 - (B) zero
 - (C) 12 N-s
 - (D) 12.5 N-s

Section – B (Maximum Marks: 12)

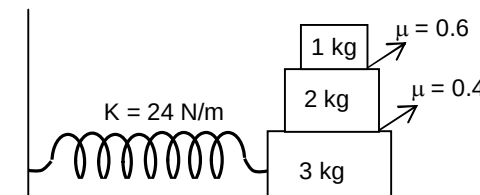
*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

11. A hydrogen-like atom of atomic number Z is in an excited state of quantum number $2n$. It can emit a maximum energy photon of 204 eV. If it makes a transition to quantum state n, a photon of energy 40.8 eV is emitted. Ground state energy of hydrogen atom is -13.6 eV. Find the value of $n + Z$.

12. A uniform solid cylinder of radius, $R = 15$ cm rolls over a horizontal plane passing into an inclined plane forming an angle $\alpha = 30^\circ$ with the horizontal as in Figure. Find the maximum integral value of the velocity v_0 (in m/s) which still permits the cylinder to roll on to the inclined plane section without a jump. The sliding is assumed to be absent.



13. What can be the maximum amplitude (in m) of the system so that there is no slipping between any of the blocks



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

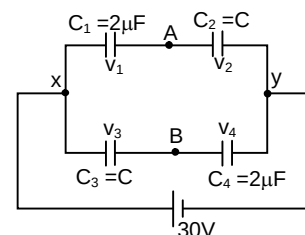
A 10 m long horizontal stainless steel wire AB of mass 1 kg whose end A is fixed, is connected to a massless string BC passing over a smooth pulley. String BC is connected to a container of mass 2 kg at end C. Water (density $= 1 \times 10^3 \text{ kg/m}^3$) is poured in the container at a constant rate of 2.25 litre/sec at $t = 0$. Also at $t = 0$, a pulse is generated at end A. Choose from the following the correct option(s). ($g = 10 \text{ m/s}^2$)

14. Find time taken by the pulse to reach point B in sec.
15. Find tension (in N) in the string at this moment.

Question Stem for Question Nos. 16 and 17

Question Stem

The given circuit shows an arrangement of four capacitors. A potential difference 30V is applied across the combination. It is observed that potentials at points 'A' and 'B' differ by 5V. Also if a conducting wire is connected between 'A' and 'B' electrons will flow from A to B. Of course, we have not connected any wire actually between A and B, we have described only an 'if' situation.



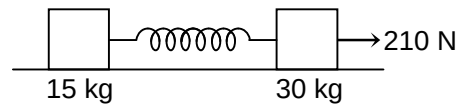
Answer the following questions:

16. Potential difference across C_4 is _____ Volts.
17. Equivalent capacitance between X and Y is _____ μF .

Question Stem for Question Nos. 18 and 19

Question Stem

In the system shown in the figure, the mass 30 kg is pulled by a force of 210 N. Answer the following questions at the instant when the 15 kg mass has acceleration 6 m/s^2 . Assume the spring to be mass less and spring constant is 100 N/m . The surface of ground is smooth.



18. Find the acceleration of 30 kg mass (in m/sec^2)
19. Find the elongation in the string at this instant (in m)

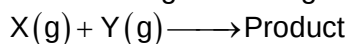
Chemistry**PART – II****Section – A (Maximum Marks: 24)**

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. The paramagnetic species out of the following is/are

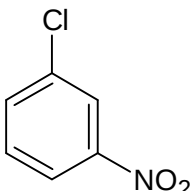
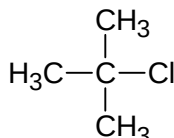
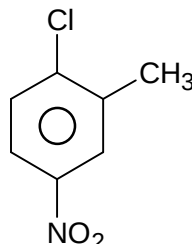
- (A) O_2^+
 (B) N_2^+
 (C) O_2^{2-}
 (D) N_2^{2-}

21. The following data are given for the reaction

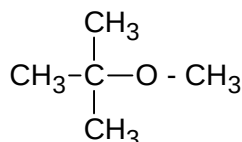


No. of expt.	Initial concentration of 'X' in 'M'	Initial concentration of 'Y' in 'M'	Rate of reaction in $M\ s^{-1}$ unit
1.	0.1	0.1	2×10^{-8}
2.	0.2	0.2	4×10^{-8}
3.	0.4	0.2	8×10^{-8}

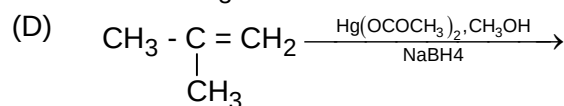
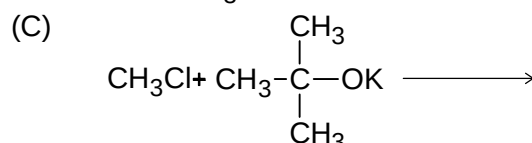
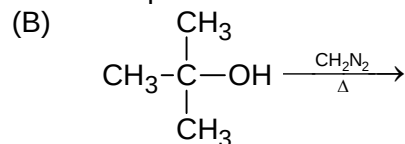
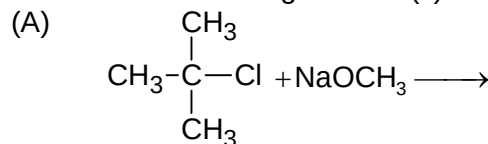
Choose the correct statement(s) from the following

- (A) The order of reaction w.r.t X is one
 (B) The overall order of reaction is one
 (C) The rate constant of the reaction is $2 \times 10^{-7}\ s^{-1}$
 (D) The half-life of 'Y' does not depend on its concentration
22. In which of the following compound(s) the C – Cl bond is shorter than that in C_2H_5Cl ?
- (A) 
- (B) $CH_2 = CH - Cl$
- (C) 
- (D) 

23.



Which of the following reaction(s) form the above compound?



24. Which of the following statements is/are true about an azeotropic mixture?

- (A) An azeotropic mixture boils at constant temperature
- (B) The composition of an azeotropic mixture changes on distillation
- (C) An azeotropic solution of two liquids has a boiling point lower than that of either of them when it shows positive deviation from the Raoult's law
- (D) An azeotropic solution of two liquids has a boiling point higher than that of either of them if it shows positive deviation from Raoult's law

25. Which of the following statements is / are correct

- (A) The coordination number of each type of ion in CsCl crystal is 8
- (B) A metal that crystallizes in bcc structure has a coordination number of 12
- (C) A unit cell of an ionic crystal shares some of its ions with other unit cells
- (D) Length of the unit cell in NaCl is 552 pm ($r_{\text{Na}^+} = 95\text{pm}; r_{\text{Cl}^-} = 181\text{pm}$)

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

For dissociation reactions, like $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$, the degree of dissociation of SO_3 can be determined by vapour density measurement of the reacting species through the following relation.

$$\alpha = \frac{D - d}{(n - 1)d}$$

The molecular mass of the reaction mixture at equilibrium is 60. Initially only SO_3 was taken for the reaction.

Answer the following questions on the basis of above write up.

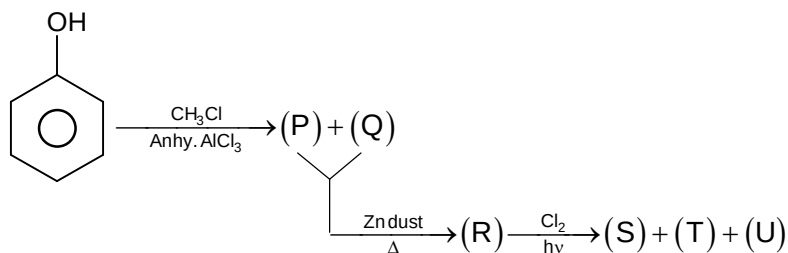
26. What is the value of 'D' in the reaction?

- (A) 60
(B) 30
(C) 40
(D) 80

27. What is the value of ' α ' for the reaction?

- (A) 0.2
(B) 0.4
(C) 0.6
(D) 0.8

Paragraph for Question Nos. 28 and 29



(S) $\rightarrow$ Monosubstituted product

(T) $\rightarrow$ Disubstituted product

(U) $\rightarrow$ Trisubstituted product

Answer the following questions on the basis of above write up.

28. The unknown compounds S, T, U are

- (A) position isomers
(B) functional isomers
(C) metamers
(D) different compounds

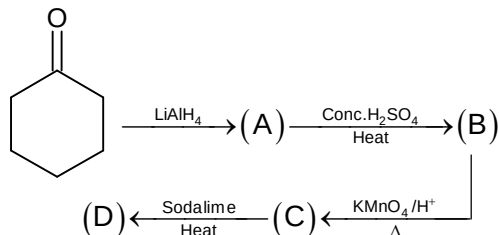
29. The most deactivating unknown compound towards EAS reaction is:

- (A) R
(B) U
(C) T
(D) S

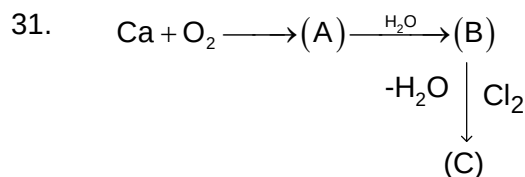
Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30.



How many maximum number of monochloro product(s) is/are formed by product(D)?



How many total numbers of atoms is/are present in (C)?

32. One mole of glycol is added to 380 g water and the solution was cooled to -5°C . How much gram of ice is formed in this process?
 [K_f of $\text{H}_2\text{O} = 1.86 \text{ K Kg mol}^{-1}$]

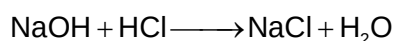
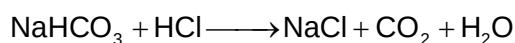
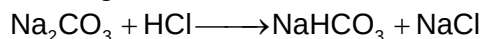
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

One litre solution contains 10.6 g of Na_2CO_3 and 4 g NaOH . Adding of HCl into the solution allows following reactions.



When the above solution mixture is treated with HCl in presence of phenolphthalein indicator, 300 mL of HCl solution is needed.

x = The concentration of HCl solution in mol L^{-1} unit

y = n-factor of Na_2CO_3 in presence of phenolphthalein indicator

33. The value of x is

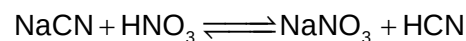
34. The value of y is

Question Stem for Question Nos. 35 and 36

Question Stem

500 mL of 0.2 M sodium cyanide(NaCN) solution was mixed with 500 mL of 0.1 M nitric acid(HNO_3) solution. The pH of the resultant solution is 'x'.

To the resulting solution, 0.01 mole of HCl was added. The pH of the solution after adding HCl is 'y'



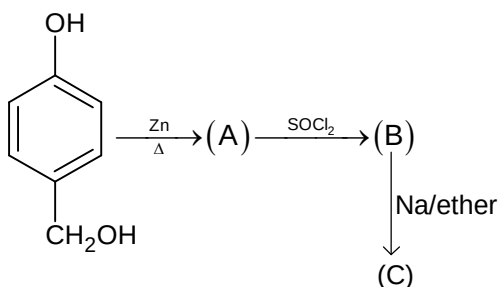
$$[\text{K}_a \text{ of } \text{HCN} = 10^{-10}] \left[\log \frac{2}{3} = -0.17 \right]$$

35. The value of x is

36. The value of y is

Question Stem for Question Nos. 37 and 38

Question Stem



x = Number of monochloro products formed by compound(C) when it reacts with $\text{Cl}_2/h\nu$

y = Number of monochloro products formed by compound(C) when it reacts with $\text{Cl}_2/\text{FeCl}_3$

37. The value of 'x' is

38. The value of 'y' is

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Let $f(x) = \begin{vmatrix} n & n+1 & n+2 \\ {}^nP_n & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^nC_n & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$, where the symbols have their usual meanings. Then

$f(x)$ is divisible by

- (A) $n^2 + n + 1$
- (B) $(n+1)!$
- (C) $n!$
- (D) none of these

40. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then

- (A) $\text{adj}(\text{adj } A) = A$
- (B) $|\text{adj}(\text{adj } A)| = 1$
- (C) $|\text{adj } A| = 1$
- (D) none of these

41. $\sin \theta + \sqrt{3} \cos \theta = 6x - x^2 - 11$, $0 \leq \theta \leq 4\pi$, $x \in \mathbb{R}$, holds for

- (A) no value of x and θ
- (B) one value of x and two values of θ
- (C) two values of x and two values of θ
- (D) two pairs of values of (x, θ)

42. Diagonals of a square are along pair of lines $2x^2 - 3xy - 2y^2 = 0$. If $(2, 1)$ is a vertex of square then sides of square can be

- (A) $3x - y + 5 = 0$
- (B) $x + 3y + 5 = 0$
- (C) $2x - y - 5 = 0$
- (D) $x - 2y = 0$

43. Let $a_n = \underbrace{(111\dots 1)}_{n \text{ times}}$, then

- (A) a_{912} is not prime
- (B) a_{951} is not prime
- (C) a_{480} is not prime
- (D) a_{91} is not prime

44. Let $f : \left[0, \frac{5}{2}\right) \rightarrow \left(\frac{1}{2}, 3\right]$ such that $f(x) = \frac{[x] + 1}{\{x\} + 1}$ where $[.]$ represents the greatest integer function and $\{.\}$ represents the fractional part of x , then which of the following is / are true.
- (A) $f(x)$ is injective discontinuous function.
 (B) $f(x)$ is surjective non – differentiable function.
 (C) $\min\left(\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x)\right) = f(1)$
 (D) $\max\left(\lim_{x \rightarrow 2^-} f(x), \lim_{x \rightarrow 2^+} f(x)\right) = f(2)$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

A sequence of ellipse $E_1, E_2, \dots, E_n$ is constructed as follows. Ellipse E_n is drawn so as to touch ellipse E_{n-1} at the extremities of the major axis of E_{n-1} and to have its foci at the extremities of the minor axis of E_{n-1} . (e_n is eccentricity of ellipse E_n)

45. If e_n is independent of n , then the eccentricity of ellipse E_{n-2} is:

- (A) $\left(\frac{3 - \sqrt{5}}{2}\right)$
 (B) $\left(\frac{\sqrt{5} - 1}{2}\right)$
 (C) $\frac{2 - \sqrt{3}}{2}$
 (D) $\frac{\sqrt{3} - 1}{2}$

46. If equation of ellipse E_1 is $\frac{x^2}{9} + \frac{y^2}{16} = 1$, then the equation of ellipse E_3 is:

- (A) $\frac{x^2}{9} + \frac{y^2}{16} = 1$
 (B) $\frac{x^2}{25} + \frac{y^2}{49} = 1$
 (C) $\frac{x^2}{25} + \frac{y^2}{41} = 1$
 (D) $\frac{x^2}{16} + \frac{y^2}{25} = 1$

Paragraph for Question Nos. 47 and 48

Let $S=0$ be the equation of the locus of point P in the plane of a square ABCD such that $\max(PA, PC) = \frac{1}{\sqrt{2}}(PB + PD)$.

47. The ratio of radii of circumcircle of square ABCD and director circle of $S=0$ is

- (A) $\frac{\sqrt{3}}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{1}{2}$
- (D) none of these

48. The ratio of perimeter of $S=0$ and the given square is

- (A) $\frac{1}{2\sqrt{2}}$
- (B) $\frac{1}{\sqrt{3}}$
- (C) $\frac{1}{7}$
- (D) $\frac{\pi}{2\sqrt{2}}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. Let z and ω be complex numbers such that $z + \omega = i$ and $z^2 + \omega^2 = 1$. If area of triangle formed by z , ω and origin is equal to A , then find the value of $16A^2$.

50. Let $\beta = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{x^2+x+1}} dx$, then find the value of β^2 .

51. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y , where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If time to empty the tank be T minutes, then $\frac{T}{10} =$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53**Question Stem**

Suppose f , g and h be three real valued functions defined on $\mathbb{R}$. Let

$$f(x) = 2x + |x|, \quad g(x) = \frac{1}{3}(2x - |x|) \text{ and } h(x) = f(g(x))$$

52. The number of integers in the range of the function

$$k(x) = 1 + \frac{1}{\pi}(\cos^{-1}(h(x)) + \cot^{-1}(h(x))) \text{ is equal to:}$$

53. The number of non negative integers in the domain of definition of the function

$$l(x) = \sin^{-1}(f(x) - g(x)) \text{ is equal to:}$$

Question Stem for Question Nos. 54 and 55**Question Stem**

A parabola touches the bisectors of the angle obtained by the lines $x + 2y + 3 = 0$ and $2x + y = 0$ at the point $M(1, 1)$ and $N(0, -2)$.

54. The absolute value of the sum of the x and y coordinates of the focus of the parabola is

55. If the equation of the directrix is given as $ax + by + 4 = 0$ then the value of $a + b$ is

Question Stem for Question Nos. 56 and 57**Question Stem**

Let a, b, c be the sides of triangle ABC such that $2b = a + c$. If $\frac{2!}{1!9!} + \frac{2!}{3!7!} + \frac{1!}{5!5!} = \frac{8^a}{(2c)!}$, then

56. Area of $\triangle ABC$ is

57. Inradius of $\triangle ABC$ is